

De novo small-molecule design campaign

[BRANDING SLOT]
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Target: glycogen synthase kinase-3 beta (GSK3B, UniProt P49841, ChEMBL CHEMBL262)

Generated 2026-10-03 · graph genetic algorithm · 8,175 molecules evaluated · 10 designs selected

8,175

unique molecules
generated & scored

7,170

valid, un-gated

248

passed full
filter cascade

10

designs selected
(distinct chemotypes)

7/10

with a proposed
synthetic route

What this report is. Ten computationally designed molecules, generated de novo and ranked by a surrogate activity model combined with drug-likeness and synthetic-accessibility heuristics. **None of these compounds has been synthesised or tested.** Every activity number here is a model prediction, not a measurement, and the model is the same kind of object the generator was optimising against - see Limitations before acting on any ranking.

1. Introduction

GSK3B is a constitutively active serine/threonine kinase that sits at the intersection of Wnt/beta-catenin signalling, insulin signalling and microtubule regulation. Its dysregulation is implicated in the principal hallmarks of Alzheimer's disease - it phosphorylates tau, modulates amyloid-beta production, and affects neurogenesis and synaptic function - which has made it a long-standing and heavily prosecuted central-nervous-system target [6]. That history cuts both ways for a design exercise: the target is well validated and richly annotated, so a data-driven generator has plenty to learn from, but the accessible chemical space around it has also been explored extensively by medicinal chemists, so genuinely novel chemotypes are correspondingly harder to reach.

This campaign asks a narrow, tractable question: starting from known potent GSK3B binders, can a graph genetic algorithm produce molecules that (a) score well on a pretrained GSK3B activity surrogate, (b) are drug-like and synthetically plausible, and (c) are structurally distinct from the compounds it started from? The deliverable is a ranked shortlist of ten designs with properties, novelty measurements, structural alerts and proposed retrosynthetic routes - a starting point for triage by a medicinal chemist, not a set of conclusions.

2. Methods

2.1 Reference and seed data

Known actives were retrieved from the ChEMBL REST API (release **ChEMBL_37**, retrieved 2026-10-03) for target CHEMBL262, restricted to IC50/Ki/Kd measurements with pChEMBL ≥ 7 (that is, 100 nM or more potent). This returned 2,195 activity records spanning **1,830 unique compounds**, each represented by its most potent measurement. This set served two distinct roles: the source of the GA's starting population, and the reference against which novelty was later measured. The starting population was the 200 most structurally diverse members, chosen by MaxMin picking over Morgan fingerprints so the GA did not begin inside a single chemotype.

2.2 Generation

Molecules were generated by a graph genetic algorithm operating directly on molecular graphs, following the operator family of Jensen [1] and the graph-GA baseline of GuacaMol [2], implemented here on RDKit rather than taken from either codebase. Crossover fragments both parents at a random acyclic single bond and re-joins one fragment from each. Mutation applies one local graph edit: atom append (from a library of common medicinal-chemistry rings and substituents), element substitution, terminal-atom deletion, bond-order change, or atom insertion into a bond. Every child is sanitised and round-tripped through SMILES; anything RDKit rejects is discarded. Selection is elitist truncation with rank-weighted parent sampling. Parameters: population 200, 40 generations, 200 offspring per generation, mutation rate 0.5, random seed 42. Wall-clock runtime was 229 s on CPU.

2.3 Objective function

Each molecule was scored as the geometric mean of three normalised terms - surrogate GSK3B activity, QED [3], and a synthetic-accessibility term $(10 - \text{SA_Score})/9$ [4]. The geometric mean is deliberately conjunctive: a zero in any term zeroes the score, so the GA cannot trade away drug-likeness for predicted potency. Activity is the pretrained GSK3B oracle distributed with the Therapeutics Data Commons [8] - a random forest over ECFP features returning a value in $[0,1]$. A hard gate additionally zeroed any molecule outside MW 150-600, 10-50 heavy atoms, $\text{cLogP} \leq 6$, $\text{SA_Score} \leq 6$, zero formal charge, single fragment, no ring larger than 7, and elements restricted to C/N/O/S/F/Cl/Br. Without this gate a GA optimising a surrogate reliably drifts into high-scoring but unmakeable space.

2.4 Filtering and selection

All molecules seen during the run - not only the final population - formed the selection pool. The cascade applied, in order: validity, uniqueness and gate survival; novelty (maximum Tanimoto < 0.40 against every one of the reference actives, Morgan radius 2, 2048 bits); activity ≥ 0.50 and QED ≥ 0.60 ; freedom from PAINS A/B/C substructures [5]; ring sanity; and $\text{SA_Score} \leq 4.5$. Selecting the ten highest-scoring survivors directly produced ten decorations of a single core, so final selection instead clustered the survivors (Butina, Tanimoto 0.4) and took the best-scoring representative of each of the ten highest-scoring clusters. Bemis-Murcko scaffold uniqueness was tried first and rejected as too weak a criterion: swapping a pendant ring changes the Murcko scaffold while leaving the recognition motif intact.

2.5 Retrosynthesis and structural alerts

Routes were proposed with AiZynthFinder [7] using the public USPTO template library, expansion and filter policies, and the ZINC stock, with a 120 s search budget per molecule. Designs were additionally screened against the Brenk reactive/unsuitable-group alerts and an explicit list of Michael-acceptor SMARTS; these were recorded as annotations rather than used as filters, because several validated GSK3B chemotypes are themselves maleimides.

3. Results

3.1 Convergence

The GA converged smoothly. Population mean score rose from 0.662 to 0.851 over 40 generations while the best individual improved from 0.874 to 0.895; the best score among molecules that were also novel by the Tanimoto < 0.4 criterion rose from 0.633 to 0.856. The best-score curve is a staircase, which is expected under elitist truncation. The mean curve flattening after roughly generation 25 indicates the population had largely converged; the run was not extended because additional generations were producing decorations of chemotypes already present rather than new ones.

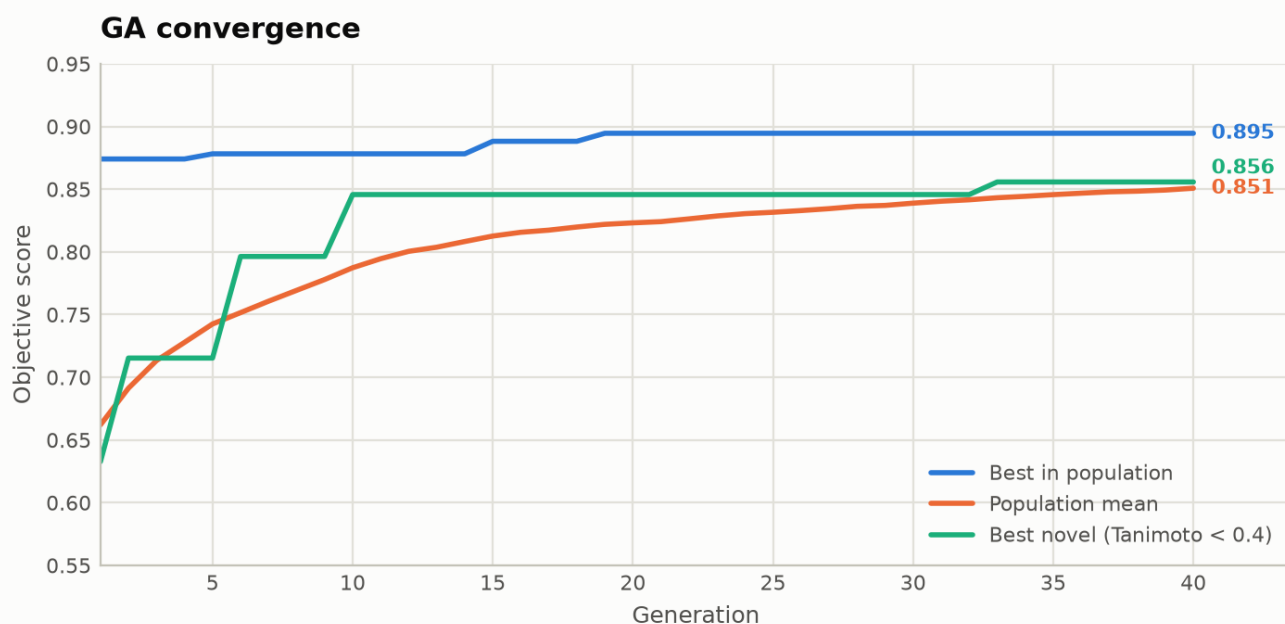


Figure 1. Objective score by generation. The novel-only curve sits persistently below the unconstrained best, which is the central tension of this campaign: the surrogate rewards resemblance to known actives, while the novelty criterion penalises it.

3.2 Filter attrition

Of 8,175 unique molecules generated, 7,170 were valid, unique and within the property gate. Novelty was by far the harshest filter, removing 4,566 molecules (64% of those entering it) - the generator spends most of its effort near its starting material. The combined activity and QED gate removed a further 2,340. PAINS, ring sanity and synthetic accessibility together removed only 16, which says less about the designs' quality than about the property gate having already excluded most of what those filters target. 248 molecules survived the full cascade.

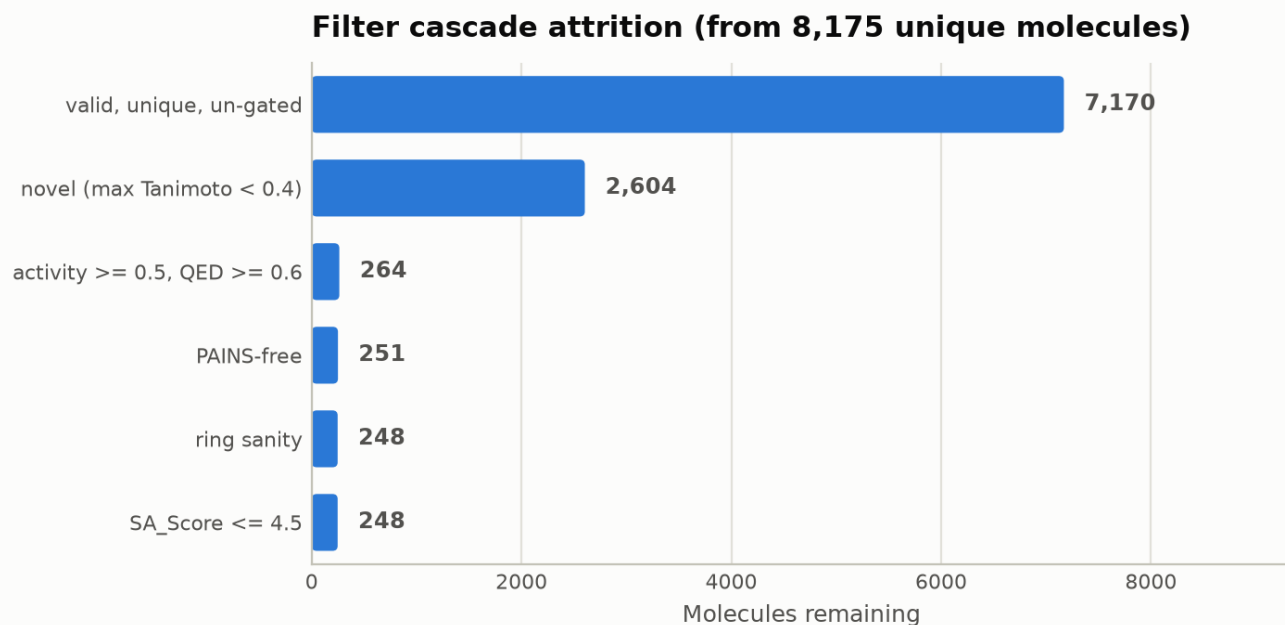


Figure 2. Molecules remaining after each cascade stage, applied in the order shown.

3.3 Novelty

Novelty here is a measured quantity, and the measurement is sobering. Across the whole generated set the modal similarity to the nearest known active is around 0.45-0.50. Among the 248 cascade survivors the distribution is compressed hard against the 0.40 cutoff: 154 of 248 sit between 0.35 and 0.40, 69 between 0.30 and 0.35, 25 between 0.20 and 0.30, and **none below 0.20**. The ten selected designs span 0.36-0.40. These molecules are

therefore novel only in the specific, threshold-relative sense defined in Methods. They are not new chemotypes; they are new points near the edge of a well-populated region. A different cutoff would have produced a different answer, and that sensitivity is itself the finding.

Similarity of generated molecules to the nearest known active

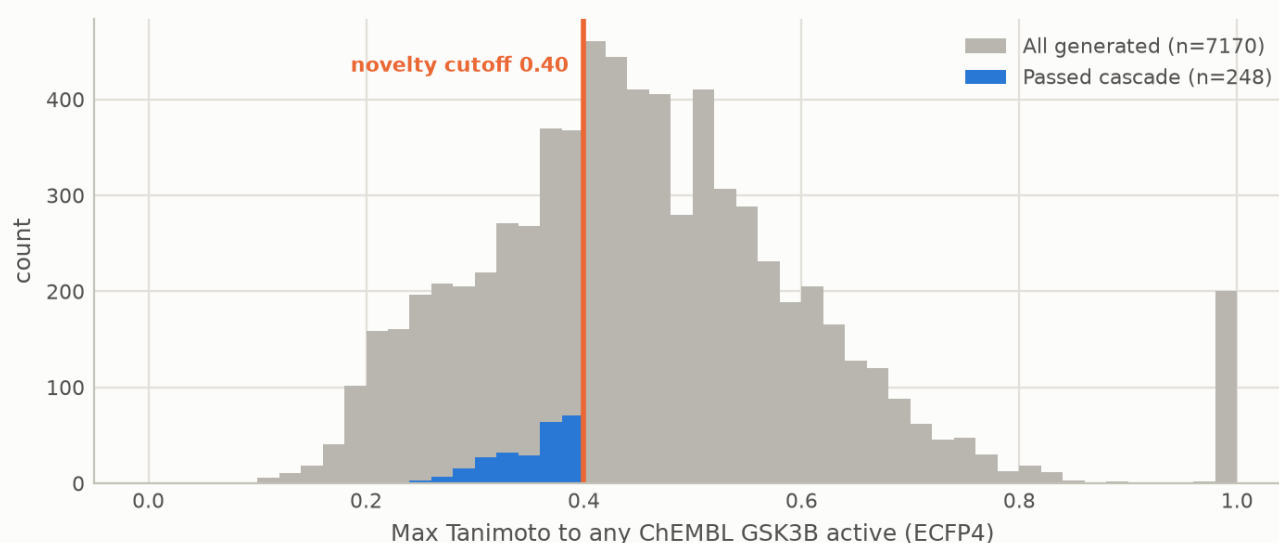


Figure 3. Maximum Tanimoto similarity of each generated molecule to the nearest of the 1,830 reference actives. The spike at 1.0 is the seed molecules, which are themselves known actives.

3.4 Property profile

The surviving designs are systematically smaller and less polar than the known actives they were derived from - the generated molecular-weight distribution centres near 250-350 Da against roughly 350-450 Da for the ChEMBL set, with correspondingly lower TPSA. This is a known bias of scoring functions containing a QED term, since QED peaks in the middle of the property ranges rather than at the upper end where many optimised kinase inhibitors sit. Synthetic accessibility overlaps well with the reference set. The practical implication is that these designs have headroom to grow during optimisation, which is a reasonable place for a starting point to be.

Property distributions: designs vs known actives

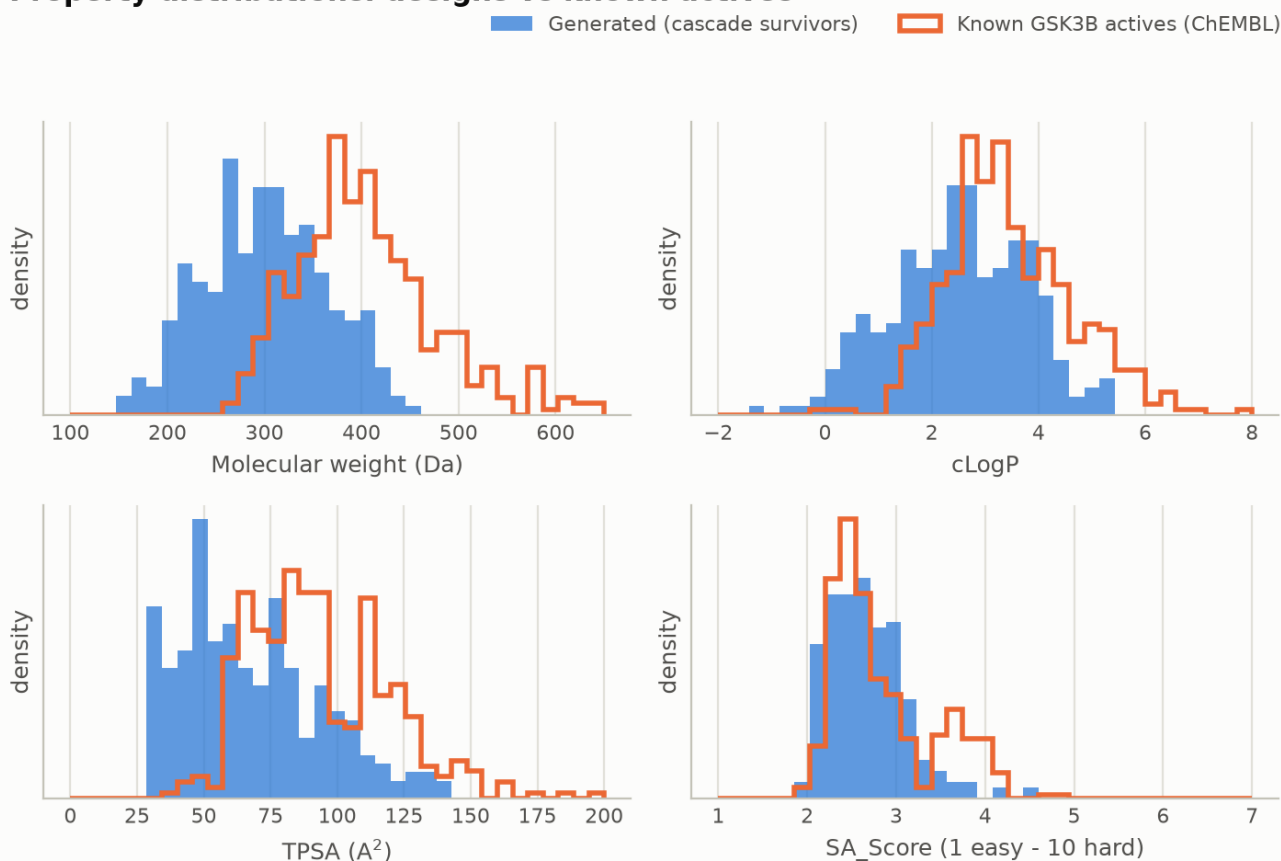


Figure 4. Property distributions for cascade survivors (filled) against a random 400-compound sample of the ChEMBL reference actives (outline).

3.5 Selected designs

Ten designs were selected as the best-scoring representatives of the ten highest-scoring chemotype clusters (of 85 clusters among the survivors). Chemotype-aware selection cost roughly 0.03 in mean objective score relative to taking the top ten outright (0.804 against 0.841) and bought a portfolio spanning recognisably different recognition motifs rather than ten variations on one.

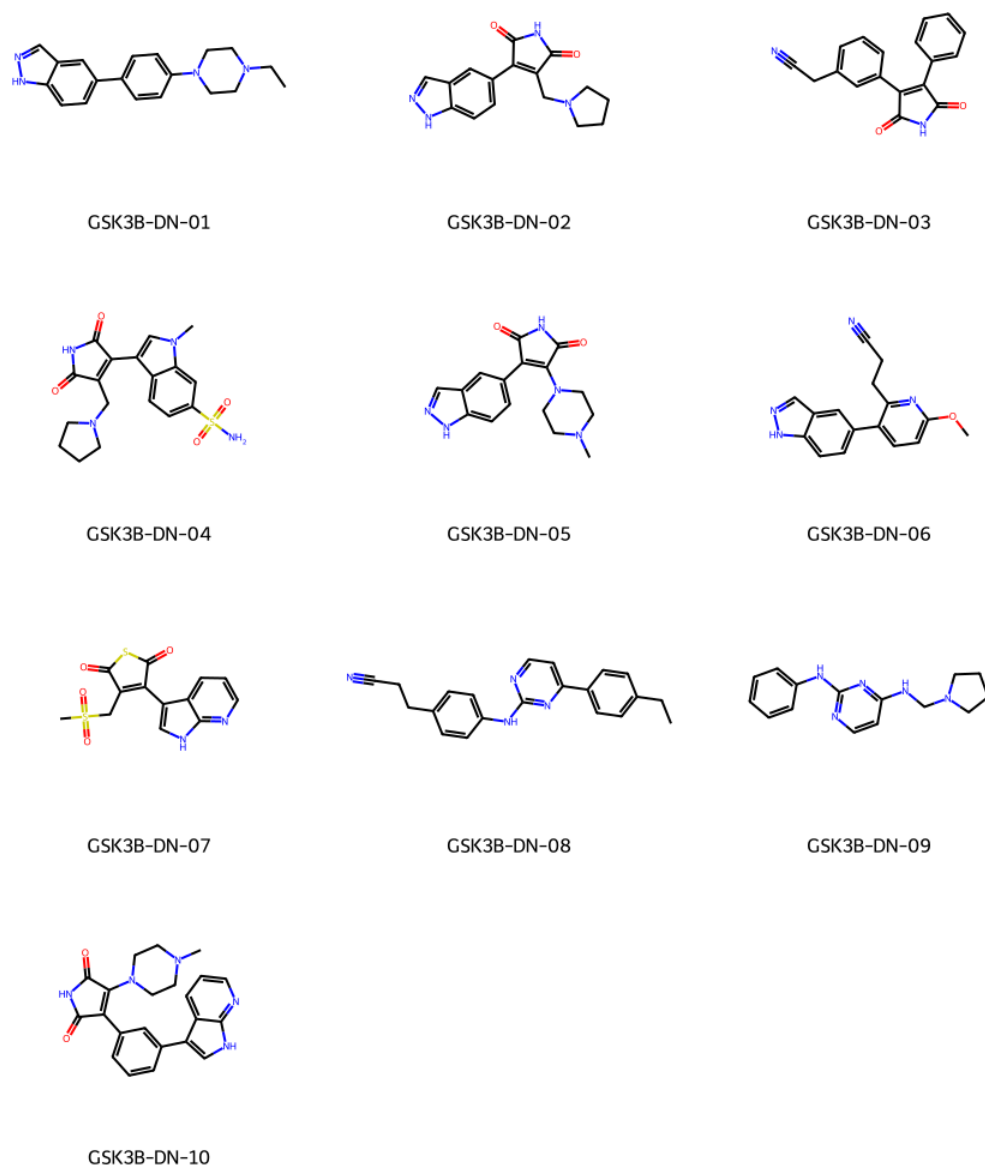


Figure 5. The ten selected designs, one per chemotype cluster. Scores, properties and alerts are in Table 1; they are surrogate-model outputs and heuristics, not measurements.

ID	Score	Act.	QED	SA	Sim.	MW	cLogP	TPSA	Route	Alerts
DN-01	0.856	0.88	0.81	2.04	0.39	306	3.4	35	1 step	-
DN-02	0.828	0.84	0.83	2.67	0.39	296	1.1	78	unsolved	maleimide, Michael
DN-03	0.810	0.70	0.88	2.25	0.40	288	2.3	70	3 steps	maleimide, Michael, nitrile
DN-04	0.804	0.88	0.73	2.76	0.38	388	0.3	114	2 steps	maleimide, Michael, sulfonamide
DN-05	0.796	0.80	0.77	2.65	0.37	311	0.2	81	4 steps	maleimide, Michael
DN-06	0.796	0.78	0.80	2.68	0.36	278	3.1	75	unsolved	nitrile
DN-07	0.793	0.72	0.91	3.15	0.39	322	1.2	97	unsolved	thiophene dione, Michael
DN-08	0.792	0.81	0.70	2.09	0.36	328	4.9	62	1 step	nitrile
DN-09	0.786	0.64	0.87	2.18	0.38	269	2.7	53	2 steps	-
DN-10	0.778	0.86	0.67	2.66	0.39	387	1.8	81	6 steps	maleimide, Michael

Table 1. Selected designs. Act. = surrogate GSK3B oracle output; Sim. = maximum Tanimoto to any reference active; Route = shortest AiZynthFinder route found within the 120 s budget. Full SMILES, scaffolds and nearest-neighbour references are in the accompanying CSV.

3.6 Retrosynthesis

AiZynthFinder returned at least one fully-solved route (all leaves in the ZINC stock) for **7 of the 10** designs, with shortest routes of 1-6 steps. Several proposals are chemically conventional and plausible on their face - DN-01 resolves to a Suzuki coupling of an aryl boronic acid with a bromo-indazole, and DN-08 to an SNAr between a methylthio-pyrimidine and an aniline. The three unsolved cases (DN-02, DN-06, DN-07) returned partial trees whose leaves were not all purchasable within the budget; that is a statement about the template library and the search budget, not proof that the molecules are inaccessible. All routes are template-based proposals and carry no information about yield, selectivity or protecting-group strategy.

4. Conclusions

The pipeline works end to end and is cheap. 8,175 molecules were generated, scored on three objectives, filtered, clustered and routed in under 6 minutes of CPU time, with a fixed seed and reproducible outputs.

Optimisation succeeded; novelty largely did not. The objective improved substantially and the designs are drug-like and synthetically tractable. But no survivor fell below Tanimoto 0.20 of a known active, and the survivors pile up immediately beneath the 0.40 cutoff. The honest reading is that the GA rediscovered and recombined established GSK3B pharmacophore families - maleimides, anilinopyrimidines, azaindoles, indazoles - rather than inventing new ones.

That rediscovery is itself a weak positive control. A generator optimising a GSK3B surrogate converging onto chemotypes that real GSK3B programmes have converged on suggests the surrogate is tracking something real. It is not evidence that the specific molecules are active.

Chemotype collapse is the default and must be designed against. The naive top ten were ten decorations of one core. Only explicit clustering produced a portfolio. Any campaign reporting a top-N list without a diversity criterion is probably reporting one bet N times.

5. Limitations

These limitations are not boilerplate; several of them materially affect how the ranking above should be read.

The activity score is a surrogate, and the generator optimised against it directly. This is the dominant caveat. The GSK3B oracle is a random forest over fingerprints trained on public data; a genetic algorithm given thousands of queries will find inputs that score highly partly because of model artefacts. High oracle output is evidence of resemblance to the model's training actives, not of binding. No docking, free-energy or structural validation was performed.

Novelty is threshold-relative and marginal. The 0.40 Tanimoto cutoff is a convention, not a property of chemistry. Survivors cluster at 0.35-0.40, so modest changes to the cutoff, the fingerprint or the radius would materially change which molecules qualify.

Six of ten designs carry structural alerts; five are maleimides. Maleimides and the thiophene-dione in DN-07 are Michael acceptors with covalent-reactivity and promiscuity liabilities. PAINS filtering does not catch them. They were retained because validated GSK3B inhibitors (SB-216763, bisindolylmaleimides) share the motif - but a maleimide-rich shortlist is a real liability, not a neutral observation, and DN-01, DN-06, DN-08 and DN-09 are the alert-free designs.

QED and SA_Score are heuristics. QED encodes an aggregate notion of drug-likeness that penalises the larger, more polar molecules common among optimised kinase inhibitors, which is visible in Figure 4. SA_Score is a fragment-frequency proxy and systematically flatters molecules built from common fragments - which is precisely what this generator builds.

Retrosynthetic routes are proposals from USPTO templates. A solved route means the search reached purchasable leaves. It implies nothing about yield, selectivity, scale or protecting groups, and the template library reflects historically published reactions.

Single run, single seed. One GA run at seed 42. No replicate runs, so the variance of the outcome is unmeasured and the specific ten molecules should be treated as one sample from a distribution, not as the optimum.

Selectivity was not assessed at all. GSK3A shares close to identical ATP-site architecture with GSK3B, and nothing in this pipeline addresses kinome-wide selectivity, cellular activity, permeability, CNS penetration or toxicity.

6. Next steps

Break the surrogate's grip. Re-score the 248 survivors with an orthogonal method the GA never saw - docking into a GSK3B ATP-site structure (several are available, e.g. 1Q5K, 6B8J) or a co-folding model - and keep only

molecules that both methods like. Disagreement between the two is the most informative signal available without experiments.

Run replicates and report variance. Five to ten seeds, then report how often each chemotype recurs. Chemotypes that survive across seeds are worth more than the single best score from one run.

Put novelty in the objective, not just the filter. Adding an explicit novelty or diversity reward would let the GA trade predicted potency for genuine structural departure, instead of being filtered after the fact - which is what produced the pile-up at the cutoff.

Triage the maleimides deliberately. Decide as a programme whether covalent/reactive chemotypes are in scope. If not, re-run with a Michael-acceptor exclusion in the gate; the alert-free designs (DN-01, DN-06, DN-08, DN-09) indicate the pipeline can produce them.

Add a GSK3A counter-screen and an ADMET pass before any synthesis decision, and have a medicinal chemist review the proposed routes - template plausibility is not a synthesis plan.

7. References

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- [10] RDKit: Open-source cheminformatics. rdkit.org

Literature references [1]-[7] were retrieved and verified via PubMed. Bioactivity data were retrieved from ChEMBL. All ten designs are computational proposals that have not been synthesised or assayed.